

Session protocol

Every session follows the same four beats. The format exists to force the study techniques that actually work — **retrieval, spacing, worked-example-then-fade, self-explanation** — and to keep each block inside 30–45 minutes.

The four beats

1. **Frame (≈ 2 min) — me.** I state what the session covers, the *one* idea to hold onto, and the exact pointers (notes subsection; book pages for depth). If a spaced-review pull is due (see below), it happens here first.
2. **Read (≈ 8 -12 min) — you.** Read the pointed chunk. My notes are the primary text; the book is the source you consult for depth. Work every derivation by hand as you read — do not let your eyes slide over a step you couldn't reproduce.
3. **Recap (≈ 8 -10 min) — you explain it back to me, notes closed.** This is retrieval practice plus self-explanation: *you* summarise and justify, *I* probe and correct on the spot. The closed notes are essential — the point is to recall, not to reread.
4. **Exercises (≈ 10 -15 min) — you attempt, I check.** One or two problems. You reason first; I debug the reasoning. Early on you study worked derivations; as you gain fluency I blank out steps ("completion problems") and then give fresh problems — the worked-example fade.

Spaced review (the every-third-session rule)

Retrieval is only half the evidence base; **spacing** is the other half. So:

- **Every third completed session** (0.3, 1.3, 2.3, ...) opens beat 1 with a **5-minute cold re-derivation** of an *earlier* result — pulled from the **Spaced-review queue** in `PROGRESS.md`, oldest un-reviewed item first.
- Do it from a blank page before checking. Then mark it reviewed and rotate it to the back of the queue. Struggling to recall is expected and is the mechanism, not a failure.
- The queue activates once there is a backlog to draw on — in practice from about session 1.3.

Difficulty flex

- **Light / reference stretches** (much of Module 0): read on your own, bring only questions — no live reading beat needed.

- **Hard machinery** (Chapters 3–5: generating functions, asymptotics, analytic combinatorics): fully interactive from beat 1; a single "reading" may span two blocks.

The principles, in one line each

- **Retrieval** — recall from blank beats rereading, every time.
- **Spacing** — return after a gap; same-day repetition barely helps.
- **Worked-example fade** — study full derivations, then complete partial ones, then solve fresh.
- **Self-explanation** — justify each step in your own words (this also trains your English).
- **Desirable difficulty** — if it feels effortful and slightly uncomfortable, it's working.